2025-Winter Paper 3 – Subspecialities of Urology

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Q1. Etiopathogenesis, diagnosis and management of Recto-Prostatic fistula

Type: LAQ (25 marks) (also answered in 2020-Summer Paper 3, Q1, as recto-urethral fistula)

Assumed meaning: rectourethral fistula at the level of the prostatic urethra or the vesicourethral anastomosis.

Definition

- **RUF (rectourethral fistula):** a communication between the prostatic urethra (or the vesicourethral anastomosis after prostatectomy) and the distal rectum.
- Rare but very morbid. Almost never seen in females, because the vagina lies between urethra and rectum.

Aetiology

Group	Causes
Congenital	Anorectal malformation in boys (rectoprostatic or rectobulbar fistula)
Surgical (commonest acquired)	**RP (radical prostatectomy)**; historically TURP (transurethral resection of the prostate) and open simple prostatectomy
Radiation and energy	**EBRT (external beam radiotherapy) plus brachytherapy** (commonest), then brachytherapy alone, then pelvic radiotherapy; cryotherapy; HIFU (high-intensity focused ultrasound): low after primary, up to about 9% after salvage HIFU
Others	Trauma (penetrating, pelvic fracture); Crohn disease; advanced rectal or prostate cancer

Pathogenesis

- **Surgical:** the rectum is torn during apical or posterior dissection behind Denonvilliers fascia, especially with small glands and salvage surgery. An unrecognised tear leaks urine, an abscess forms and the tract epithelialises. Rectal injury is commonest with retropubic and laparoscopic RP, lowest with robotic RP, and was over 15% in historical open salvage series.
- **Radiation or ablation:** obliterative endarteritis makes the tissue hypoxic and fibrotic. Necrosis follows months to years later. These fistulae are large and heal poorly.

Classification: Martini (after prostate cancer treatment)

Axis	Category	Definition
Stage (size on CT (computed tomography) urography or cystography)	I / II / III	≤1.5 cm / >1.5 cm / any size with sphincter damage
Grade (tissue)	0	Surgical injury, no prior non-surgical treatment
	1	Prior radiation, brachytherapy, HIFU or other energy
	2	Salvage RP or salvage ablation

- **Tissue quality (grade), not size, chiefly decides the repair.**

Clinical features

- **Faecaluria and pneumaturia** (most typical); urine per rectum on voiding.
- Recurrent UTI (urinary tract infection), dysuria, urosepsis; coexisting SUI (stress urinary incontinence).

Diagnosis

- **DRE (digital rectal examination):** may feel the defect on the anterior rectal wall.
- **Cystoscopy and proctosigmoidoscopy:** show the tract and tissue vitality; **biopsy** to exclude recurrent cancer.
- **VCUG (voiding cystourethrogram) or RUG (retrograde urethrogram):** usually conclusive; gives size and site for planning.
- Upper-tract imaging to exclude ureteric injury; pelvic MRI (magnetic resonance imaging) for complex cases.
- **UDS (urodynamics)** with rectal packing: repair alone will not cure severe SUI; a later AUS (artificial urinary sphincter) erodes more often.

Prevention

- At RP: a recognised tear without gross spillage → multilayer closure, air test, **omental or peritoneal interposition**, drain; colostomy if previously irradiated.
- Radiotherapy: a **hydrogel rectal spacer** lowers rectal dose and late rectal toxicity.

Management

- **Principles:** control sepsis; divert urine ± faeces; let inflammation settle (about 12 weeks); exclude cancer; then excise the tract and close urethra and rectum separately in **tension-free, non-overlapping layers with vascularised tissue interposed**.
- **Conservative:** no sepsis and no faecaluria → catheter (urethral or suprapubic), low-residue diet ± colostomy. A simple fistula may close in about 12 weeks; irradiated or complex ones rarely do.
- **Surgery:** no single accepted operation; choice depends on site, radiation and prior repair. **A diverting colostomy usually comes first.**

Approach	Essentials	Strengths	Weaknesses
Perineal (Young; Goodwin)	Close urethra and rectum separately; **gracilis flap** between (or dartos; buccal graft)	High closure; most stomas reversed	Urinary incontinence common
Transsphincteric (York-Mason)	Prone jackknife; sphincter divided in tagged layers; through anterior rectal wall; three-layer closure	Avoids scarred plane; spares sexual function	Poor membranous exposure; no flap; faecal incontinence up to 20%

Approach	Essentials	Strengths	Weaknesses
Abdominal (transvesical, extravesical, O'Conor-like)	Omental flap	Large or irradiated fistulae, extensive resection	Morbid; poor deep-pelvic view
Laparoscopic / robotic	Open pouch of Douglas; close rectum in two layers; omental or peritoneal flap; can combine with perineal gracilis	Deep-pelvic vision	Small series only
Salvage	Cystoprostatectomy, ileal conduit ± permanent colostomy	Unreconstructable irradiated field	Major surgery

```
RUF → sepsis control, catheter ± colostomy, biopsy
  small, non-irradiated, no faecaluria → conservative ~12 weeks
  persistent, grade 0 → perineal (gracilis) or York-Mason
  irradiated or large → abdominal/robotic (omentum) or perineal (gracilis)
  unreconstructable → urinary + faecal diversion
Congenital → newborn colostomy → posterior sagittal anorectoplasty
```

- **PSARP (posterior sagittal anorectoplasty)** separates rectum from urethra and places it within the sphincter complex.

Complications

- **Recurrence** (the greatest challenge), urinary or faecal incontinence, urethral stricture, ED (erectile dysfunction), wound infection, permanent stoma.

Diagram to draw: sagittal pelvis with the fistula and a gracilis flap between urethra and rectum.

> **Key points to remember:**
> - Most acquired RUFs follow prostate cancer treatment.
> - Martini: stage = size; grade = tissue (surgery, radiation, salvage).
> - VCUG or RUG plus endoscopic biopsy to exclude cancer.
> - Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

Q2. Diagnosis and Management of Priapism

Type: LAQ (25 marks) (also answered as an SAQ in 2015-Summer Paper 2, Q3a)

Definition

- Full or partial erection lasting **≥4 hours** beyond sexual stimulation, or unrelated to it (AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2021–2022).

Classification

	Ischaemic (low flow)	Stuttering (recurrent ischaemic)	Non-ischaemic (high flow)
Share	**95%**	Subset of ischaemic	<5%
Lesion	Veno-occlusion	Repeated ischaemic episodes	Cavernosal artery–sinusoid fistula
Pain, rigidity	Painful, fully rigid	Painful, recurrent, often nocturnal	Painless, partial
Typical cause	Drugs, SCD (sickle cell disease)	SCD	Perineal or needle trauma
Urgency	**Emergency**	Emergency per episode + prophylaxis	Not an emergency

Aetiology

- **Ischaemic:** ICI (intracavernosal injection: papaverine, phentolamine, PGE1 (prostaglandin E1)), including recreational use; α-blockers, trazodone, antipsychotics, cocaine; heparin or warfarin withdrawal; **SCD** (29–42% of affected men in their lifetime; 63% of paediatric ischaemic priapism); leukaemia (1–5%, high white cell count); penile metastases; scorpion and spider venom.
- **Non-ischaemic:** straddle or coital injury, pelvic fracture, needle injury. Onset is often delayed, when the clot over the arterial tear lyses.

Pathophysiology

- **Ischaemic:** failed outflow acts like a compartment syndrome. Hypoxia, hypercarbia and acidosis cause sinusoidal thrombosis, smooth-muscle necrosis and TGF-β (transforming growth factor beta)-driven fibrosis, so ED (erectile dysfunction) follows. Irreversible damage starts within hours.
- **Stuttering (molecular):** low basal NO (nitric oxide) from eNOS (endothelial NO synthase) dysfunction downregulates PDE5 (phosphodiesterase 5). When an erection starts, cGMP (cyclic guanosine monophosphate) is not broken down, so relaxation is unopposed. Low Rho-kinase and high adenosine add to this. This explains why daily PDE5 inhibitors prevent episodes.
- **Non-ischaemic:** unregulated, well-oxygenated arterial inflow, so fibrosis is unusual.

Diagnosis

- **History:** duration, pain, drugs, trauma, SCD or blood disease, prior episodes, baseline erectile function.
- **Examination:** rigid tender corpora with a soft glans and spongiosum (ischaemic); partial, non-tender (high flow); perineal bruising; vital signs before phenylephrine.
- **Corporal blood gas** (cornerstone):

	Colour	pO2 (mmHg)	pCO2 (mmHg)	pH
Ischaemic	Dark red	**<30**	**>60**	**<7.25**
Flaccid (mixed venous)	Medium red	40	50	7.35
Non-ischaemic	Bright red	>90	<40	7.40

- **PDUS (penile duplex ultrasound)** when the gas is equivocal: ischaemic = absent cavernosal flow or PSV (peak systolic velocity) <50 cm/s with no diastolic flow; high flow = PSV >50 cm/s, may show the fistula. Angiography only for embolisation.
- Blood count, reticulocytes, haemoglobin electrophoresis, coagulation, toxicology: in parallel, never delaying treatment.

Management of ischaemic priapism (AUA/SMSNA 2021)

```
Counsel on ED risk → dorsal penile block
<36 h:
  aspiration (19 G, penoscrotal junction, 3 or 9 o'clock) ± saline irrigation
  + phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to 1 h
    (monitor blood pressure, pulse: reflex bradycardia)
  → fails → distal shunt ± tunnelling → proximal shunt or penoscrotal decompression
>36 h or refractory → early penile prosthesis (or analgesia and counselling)
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- Aspiration alone succeeds in under 1 in 4. Cold (10 °C) saline irrigation was better than body temperature in an RCT (randomised controlled trial); avoid chilled saline in SCD.
- Ice, oral agents and IV (intravenous) fluids are adjuncts only; in SCD, intracavernosal treatment still comes first.

Shunt	Method
Winter (distal)	Tru-Cut needle through glans; least effective
Ebbehoj / T-shunt (distal)	Scalpel through glans; T-shunt rotates blade 90°
Al-Ghorab (distal, open)	Dorsal glans incision; excise corporal tips
Snake tunnelling	7/8 Hegar dilator passed proximally through distal shunt
Quackels / Sacher (proximal)	Cavernosum to spongiosum at bulb; stagger ≥1 cm
Grayhack / Barry	Cavernosum to saphenous / dorsal vein
Penoscrotal decompression	Proximal corporotomy, clot suction; allows prosthesis

- **ED by duration** (Bennett and Mulhall, SCD): ≤12 h 0%; 12–24 h 22%; 24–36 h 56%; ≥36 h 100%.
- **Early prosthesis** (Zacharakis 2014): shortening 3% vs 40%, revision 9% vs 27%, satisfaction 96% vs 60% compared with delayed insertion.

Stuttering priapism

- An episode lasting ≥4 hours is treated as ischaemic; home self-injection of phenylephrine.
- **Prophylaxis:** daily PDE5 inhibitor (Burnett 2014 RCT: primary endpoint missed, but promising), pseudoephedrine, etilefrine. **Also know:** GnRH (gonadotropin-releasing hormone) agonists, antiandrogens, finasteride (cost: libido); digoxin; baclofen.

Non-ischaemic priapism (AUA/SMSNA 2022)

- Observation, perineal compression and ice; about 60% resolve. Reassess at 4 weeks.
- Persistent → PDUS → **selective cavernosal artery embolisation** (resolution ≥75%, recurrence 30–40%). For recurrence, repeat embolisation with non-resorbable PVA (polyvinyl alcohol), not surgery. Surgical ligation is a last resort (ED about 50%).

Complications

- ED, fibrosis, shortening; shunt oedema, haematoma, urethral injury or fistula, necrosis, infection.

Diagram to draw: distal (glans) and proximal shunt sites on a penile outline.

Key points to remember:

- Blood gas decides: ischaemic pO2 <30, pCO2 >60, pH <7.25.
- Aspiration plus phenylephrine first; distal shunt ± tunnelling next.
- ED risk 0% ≤12 h, 100% ≥36 h; early prosthesis beyond 36 h.
- High flow: observe, then embolise; repeat embolisation rather than surgery.

Q3a. Urethrovaginal fistula

Type: SAQ (10 marks)

Aetiology

- **Developing world:** obstructed labour, often with a VVF (vesicovaginal fistula); also trauma, radiation, catheter injury.
- **Developed world (iatrogenic): urethral diverticulectomy** (fistula in 1–8%; causes 20–25% of UVFs (urethrovaginal fistulae)), slings, bulking agents, anterior repair or mesh; also pessaries, indwelling catheters (pressure necrosis), gender-affirming and paediatric genital surgery.

Clinical features

- Depend on site relative to the sphincter: **distal** often asymptomatic; **proximal** (bladder neck) causes continuous leakage or SUI (stress urinary incontinence).
- Split stream, vaginal voiding, post-void dribble; dyspareunia; recurrent UTI (urinary tract infection).

Diagnosis

- **Speculum examination** usually suffices; **methylene blue test** if rugae hide it. Assess prolapse, SUI, atrophy.
- **VCUG (voiding cystourethrogram), lateral view**; VUDS (videourodynamics) for associated incontinence.
- **Cystourethroscopy** (flexible scope distends the urethra better): size, site, exposed mesh, other fistulae. **About 20% have a separate VVF.** Image the upper tract.
- Goh classification (distance from external meatus: type 1 >3.5 cm to type 4 <1.5 cm): closer means worse closure and continence.

Management: transvaginal rotational-flap repair

1. Lithotomy (prone gives better anterior exposure); cystourethroscopy; cannulate tract; catheter ± suprapubic.
2. Inverted **J- or U-shaped vaginal flap**; circumscribe tract with minimal excision.
3. Mobilise periurethral tissue widely.
4. **Close urethra** over a sound with 4-0 absorbable sutures; **second layer perpendicular**; buccal graft or flap onlay for large defects; leak test.
5. **Interposition: Martius flap** most often (omentum, gracilis or rectus fascia occasionally).
6. **Autologous pubovaginal sling** if SUI or bladder-neck involvement; **never synthetic**. Remove any mesh in the field.
7. Advance vaginal flap; suprapubic catheter ≥14 days; VCUG before removal; nothing per vagina for 6 weeks.

- **Near-total urethral loss:** tubularised anterior bladder flap neourethra.

Outcomes

- Closure **90–95%**, but continence only **50–80%**; postoperative SUI in up to half. Stricture may follow.

Diagram to draw: inverted-U vaginal flap with layered closure and Martius flap.

Key points to remember:

- Commonest Western cause is diverticulectomy; obstetric in developing countries.
- Look for a coexisting VVF (about 20%).
- Layered closure plus Martius flap; autologous sling only.
- Closure is easy, continence is hard (50–80%).

Q3b. Management of newly Metastatic prostate cancer

Type: SAQ (10 marks)

Definitions

- **mHSPC (metastatic hormone-sensitive prostate cancer):** no prior hormones. **Synchronous** (at diagnosis) or **metachronous** (after local therapy).
- **CHAARTED high volume:** ≥4 bone metastases with ≥1 outside the axial skeleton, or any visceral metastasis.
- **LATITUDE high risk:** ≥2 of Gleason ≥8, ≥3 bone lesions, visceral metastasis.

Work-up

- CT (computed tomography) and bone scan; PSMA (prostate-specific membrane antigen) PET (positron emission tomography) finds more, but trials used conventional imaging.
- **Germline and somatic testing** for all (>10% have DNA-repair mutations).

Systemic therapy

- **ADT (androgen deprivation therapy) is the backbone, never alone:** LHRH (luteinising hormone-releasing hormone) agonist with antiandrogen flare cover, or antagonist/orchidectomy (for impending cord compression).
- **Doublets:** ADT plus one of:

Agent	Trial	Overall survival HR (hazard ratio)
Docetaxel 75 mg/m², 6 cycles	**CHAARTED** (benefit in high volume only); STAMPEDE	0.61; 0.78
Abiraterone 1000 mg + prednisone 5 mg	**LATITUDE**; STAMPEDE arm G	0.62; 0.63
Enzalutamide	ARCHES (radiographic PFS (progression-free survival)); ENZAMET	0.67 (ENZAMET)
Apalutamide	**TITAN**	0.67

- Hormonal agents work **regardless of volume**; docetaxel needs high volume (GETUG-AFU 15 negative).
- **Triplets** (ADT + docetaxel + abiraterone or darolutamide): **PEACE-1** (abiraterone, synchronous; overall survival HR 0.75 with docetaxel) and **ARASENS** (darolutamide; overall survival HR 0.68). For chemotherapy-fit men, especially high volume.

Local and directed therapy

- **RT (radiotherapy) to the prostate in low-volume synchronous disease:** STAMPEDE arm H, survival HR 0.68 (3-year 81% vs 73%); HORRAD negative overall.
- Cytoreductive prostatectomy: trials only. **MDT (metastasis-directed therapy)** for oligometastases (STOMP, ORIOLE) delays ADT.

Follow-up

- PSA (prostate-specific antigen) and bloods 3–6-monthly; testosterone if PSA rises (rising PSA with castrate testosterone = CRPC (castration-resistant prostate cancer)); bone and cardiovascular health.
- *Guideline note:* the chapter lists ADT plus docetaxel as an option; current EAU (European Association of Urology) guidance gives docetaxel only within a triplet.

Diagram to draw: algorithm splitting by volume and fitness to doublet or triplet ± prostate RT.

> **Key points to remember:**
> - ADT alone is no longer standard.
> - CHAARTED volume vs LATITUDE risk definitions.
> - Triplets: PEACE-1, ARASENS.
> - Prostate RT only for low-volume synchronous disease (STAMPEDE arm H).

Q3c. Renal Autotransplantation

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 3, Q3c and 2021-Summer Paper 3, Q3e)

Definition and principle

- Removing a kidney and re-implanting it, usually in the **ipsilateral iliac fossa** on the iliac vessels, with **bench (ex vivo) surgery** if needed. First done in **1963** for proximal ureteric injury.
- Bringing the kidney to the bladder replaces the ureter without bowel.

Indications

Group	Examples
Ureteric loss (commonest)	Long iatrogenic or traumatic loss; repeated failed repairs (**delayed setting only, never acute**)
RPF (retroperitoneal fibrosis)	Ureterolysis impossible, function satisfactory
Tumour	Total ureterectomy for UTUC (upper tract urothelial carcinoma) in a solitary kidney; bench partial nephrectomy
Vascular	Complex branch renal artery aneurysm, stenosis or AVM (arteriovenous malformation) unsuitable for endovascular repair; nutcracker syndrome
Pain	Selected loin pain–haematuria syndrome

- **Preferred over ileal ureter** when the other kidney is absent or poor, or bowel is unsuitable (creatinine >2 mg/dL, inflammatory bowel disease, radiation enteritis).
- **Contraindications:** poor renal function, severe iliac atherosclerosis, unfit patient.

Procedure

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CT (computed tomography) angiography, renogram
→ harvest as for live donation (open, laparoscopic, robotic):
  maximal vessel length, preserve ureteric blood supply
→ cold perfusion ± bench repair
→ Gibson incision, extraperitoneal iliac fossa
→ renal vein to external iliac vein (end-to-side);
  artery to external iliac (end-to-side) or internal iliac (end-to-end)
→ pyelovesicostomy (± Boari flap), ureteroneocystostomy,
  or native ureter left in continuity
→ stent, catheter, drain
```

- Laparoscopic harvest: faster recovery; **robotic autotransplantation** emerging.

Complications and outcome

- Arterial or venous thrombosis, bleeding, urine leak, stricture, lymphocele, infection.
- **Renal unit loss 5–10%** (lower in transplant centres); major (Clavien–Dindo ≥III) complications about 15–20%.
- **Long-term success about 90%.** Doppler ultrasound early; renogram and blood pressure later.

Also know: alternatives: Boari flap, TUU (transureteroureterostomy), ileal ureter, Yang–Monti, buccal ureteroplasty, renal descensus.

Diagram to draw: kidney in the iliac fossa with vascular anastomoses and pyelovesicostomy.

Key points to remember:

- For long ureteric loss when the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for complex vascular or tumour cases.
- Unit loss 5–10%; success about 90%.

Q3d. Laparoscopic Trans vesical Prostatectomy

Type: SAQ (10 marks)

Assumed meaning: laparoscopic transvesical (suprapubic) simple prostatectomy for large BPH (benign prostatic hyperplasia).

Introduction

- Laparoscopic version of the Freyer suprapubic operation: the adenoma is enucleated through a **cystotomy** in the **surgical capsule plane** (compressed peripheral zone). First LSP (laparoscopic simple prostatectomy): Mariano, 2002.

Indications

- Gland **>80–100 g** with refractory LUTS (lower urinary tract symptoms), retention, recurrent UTI (urinary tract infection), haematuria or obstructive nephropathy.
- **Transvesical route preferred** for a large intravesical median lobe, bladder stones or a diverticulum needing surgery.
- Work-up: PSA (prostate-specific antigen) ± MRI (magnetic resonance imaging) to exclude cancer; TRUS (transrectal ultrasound) volume; cystoscopy.

Steps

1. General anaesthesia, supine, Trendelenburg. **Ports:** 10-mm umbilical camera, two 10-mm working ports, 5-mm and 12-mm assistant ports. Transperitoneal or extraperitoneal (Retzius).
2. **Cystotomy** with stay sutures; identify ureteric orifices.
3. **Circumferential bladder-neck incision** (start 3 to 9 o'clock posteriorly), away from orifices.
4. Develop the **posterior plane first** to the apex, then the lateral lobes.
5. Divide anterior commissure; **transect urethra just proximal to the verumontanum** (protects sphincter).
6. **Figure-of-eight sutures at 5 and 7 o'clock** (prostatic arteries enter posterolaterally).
7. **Bladder-neck mucosal advancement** to urethral mucosa covers the fossa and reduces bleeding.
8. 22 Fr three-way catheter; two-layer cystotomy closure.

Outcomes versus open

- Less blood loss, shorter catheter time and stay, less analgesia; **same functional results** (IPSS (International Prostate Symptom Score), flow rate, residual).
- Limitations: steep learning curve, poor ergonomics; **robotic simple prostatectomy** has largely replaced it. **Single-port transvesical robotic enucleation** (2.7 cm suprapubic incision, pneumovesicum) allows same-day discharge.

Complications

- Bleeding (transfusion rare), urine leak, ureteric orifice injury, bladder neck contracture (<1%; open 2–6%).
- **Retrograde ejaculation 80–90%**; ED (erectile dysfunction) 3–5%; incontinence rare.

Diagram to draw: sagittal view of cystotomy, bladder-neck incision and enucleation plane.

Key points to remember:

- Indication: gland >80 g, especially with median lobe, stones or diverticulum.
- Posterior plane first; divide urethra proximal to verumontanum.
- 5 and 7 o'clock sutures plus mucosal advancement.
- Same function as open, less morbidity.

Q3e. Sacral Neuromodulation

Type: SAQ (10 marks) (also answered in 2017-Winter Paper 3, Q3a)

Definition and mechanism

- **SNM (sacral neuromodulation)** (Tanagho and Schmidt): continuous low-amplitude **S3** stimulation through a tined lead and IPG (implantable pulse generator).
- It acts on **afferents** to the forebrain, not motor nerves. In OAB (overactive bladder) it damps PMC (pontine micturition centre) firing; in retention it restores filling awareness, so the PMC switches on and the sphincter relaxes. Hence it treats **both**.

Indications (third line after behavioural and drug therapy; AUA (American Urological Association) and EAU (European Association of Urology))

FDA (Food and Drug Administration) approved	Not approved
Refractory UUI (urgency urinary incontinence)	Bladder pain syndrome
Urgency–frequency	Neurogenic bladder (low-level evidence)
Non-obstructive urinary retention (Fowler syndrome responds well)	Pelvic pain
Faecal incontinence	

Technique

Test	PNE (percutaneous nerve evaluation)	Staged implant
Lead	Temporary, office, local anaesthesia	Permanent tined lead, theatre, fluoroscopy
Duration	5–7 days	Up to 4 weeks
Use	OAB	Higher conversion; retention

- Prone; S3 about **9 cm above the coccyx tip, 1–2 cm lateral**.

Root	Pelvic response	Foot response
S2	—	Calf and foot external rotation
S3	**Bellows**, perineal sensation	**Great-toe flexion**
S4	Bellows only	None

- **Implant if ≥50% improvement** in a diary parameter (retention: ≥50% fewer catheterisations); IPG in the buttock; 10–16 Hz at sensory threshold.

Outcomes

- **InSite**: success 61% vs 42% with standard medical therapy at 6 months. About half keep benefit at ≥5 years.
- **ROSETTA**: onabotulinumtoxinA 200 U slightly better for UUI (−3.9 vs −3.3 episodes/day), but more UTI (urinary tract infection) and self-catheterisation; SNM removal 9% at 2 years.

Complications and contraindications

- Pain at IPG site, lead migration, loss of efficacy, infection; costly.
- Contraindications: failed test, inability to use programmer.

Recent advances

- Rechargeable IPGs (≥15 years); **whole-body MRI (magnetic resonance imaging)-conditional** systems; bilateral no better. **Also know:** PTNS (percutaneous tibial nerve stimulation).

Diagram to draw: lead in S3 foramen with IPG in the buttock.

> **Key points to remember:**
>
> - Afferent modulation treats both OAB and retention.
> - S3: bellows plus great-toe flexion.
> - Implant if test improves ≥50%.

Q3f. Minimal invasive retroperitoneal lymph node dissection

Type: SAQ (10 marks) (cross-referenced from 2025-Winter Paper 4, Q3e)

Introduction

- **MI-RPLND (minimally invasive retroperitoneal lymph node dissection)** for testicular GCT (germ cell tumour): laparoscopic (1992) or robotic (2006). **It must duplicate open RPLND exactly.**

Indications

- **Primary:** marker-negative stage I NSGCT (non-seminomatous GCT), especially high-risk IB; small-volume stage II NSGCT or seminoma (1–2 nodes ≤3 cm in the landing zone).
- **Postchemotherapy:** only a small solitary residual mass with little desmoplasia; **open** for large masses or vessel involvement.

Templates and nerve sparing

- Boundaries: renal hila, ureters, ureter crossing the common iliacs; plus spermatic cord.
- **Primary:** extended **unilateral modified template** with nerve sparing (right: paracaval, precaval, interaortocaval; left: para-aortic, preaortic; both take nodes above the IMA (inferior mesenteric artery)). Enlarged nodes → bilateral.
- **Postchemotherapy:** bilateral full template.
- **Nerve sparing:** identify the sympathetic chains and L1–L4 postganglionic fibres to the hypogastric plexus below the IMA.

Steps (robotic)

1. Positioning: **modified lateral decubitus** (unilateral) or **supine steep Trendelenburg with peritoneal hammock** (any template).
2. Mobilise colon; Kocherise duodenum (right); divide gonadal vein.
3. **Split-and-roll:** split over the great vessel from common iliac to renal vein; roll packets off.
4. **Divide lumbar vessels** to clear retrocaval and retroaortic tissue.
5. Clip lymphatics (left renal vein); bag each packet.

Feature	Laparoscopic	Robotic
Retraction	Assistant	Fourth arm
Instruments	Straight	Wristed
Suturing, vessel control	Difficult	Easier

Outcomes

- Stay 1–3 days, blood loss mostly ≤200 mL; conversion about 5%; **antegrade ejaculation ≥90%**; chylous ascites as open. Node yields approach open (about 30).
- **Major bleeding:** pressure, raise pneumoperitoneum, convert if uncontrolled (open set ready).
- Concerns: staging-only dissections, frequent adjuvant chemotherapy, unindicated surgery, catastrophic bleeding, follow-up <5 years; no randomised trials.
- **Guidelines:** AUA (American Urological Association): experienced surgeons may offer it; EAU (European Association of Urology): specialist centres only.

Also know: serum miR-371a-3p for selection; RPLND for stage II seminoma (SEMS, PRIMETEST).

Diagram to draw: right and left modified templates.

> **Key points to remember:**
> - Same template, split-and-roll, lumbar division and nerve sparing as open.
> - Best for chemotherapy-naive low-volume disease.
> - Conversion about 5%; ejaculation preserved ≥90%.

Rapid Revision Sheet – 2025-Winter Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Etiopathogenesis, diagnosis and management of Recto-Prostatic fistula (LAQ)

- Most acquired RUFs follow prostate cancer treatment.
- Martini: stage = size; grade = tissue (surgery, radiation, salvage).
- VCUG or RUG plus endoscopic biopsy to exclude cancer.
- Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

Q2. Diagnosis and Management of Priapism (LAQ)

- 36 h or refractory → early penile prosthesis (or analgesia and counselling)
- Blood gas decides: ischaemic pO2 <30, pCO2 >60, pH <7.25.
- Aspiration plus phenylephrine first; distal shunt ± tunnelling next.
- ED risk 0% ≤12 h, 100% ≥36 h; early prosthesis beyond 36 h.
- High flow: observe, then embolise; repeat embolisation rather than surgery.

Q3a. Urethrovaginal fistula (SAQ)

- Commonest Western cause is diverticulectomy; obstetric in developing countries.
- Look for a coexisting VVF (about 20%).
- Layered closure plus Martius flap; autologous sling only.
- Closure is easy, continence is hard (50–80%).

Q3b. Management of newly Metastatic prostate cancer (SAQ)

- ADT alone is no longer standard.
- CHAARTED volume vs LATITUDE risk definitions.
- Triplets: PEACE-1, ARASENS.
- Prostate RT only for low-volume synchronous disease (STAMPEDE arm H).

Q3c. Renal Autotransplantation (SAQ)

- For long ureteric loss when the other kidney is poor or bowel unsuitable.
- Delayed setting only; bench surgery for complex vascular or tumour cases.
- Unit loss 5–10%; success about 90%.

Q3d. Laparoscopic Trans vesical Prostatectomy (SAQ)

- Indication: gland >80 g, especially with median lobe, stones or diverticulum.
- Posterior plane first; divide urethra proximal to verumontanum.
- 5 and 7 o'clock sutures plus mucosal advancement.
- Same function as open, less morbidity.

Q3e. Sacral Neuromodulation (SAQ)

- Afferent modulation treats both OAB and retention.
- S3: bellows plus great-toe flexion.
- Implant if test improves ≥50%.

Q3f. Minimal invasive retroperitoneal lymph node dissection (SAQ)

- Same template, split-and-roll, lumbar division and nerve sparing as open.
- Best for chemotherapy-naive low-volume disease.
- Conversion about 5%; ejaculation preserved ≥90%.